Lecture 8: Valuation of Bonds with Embedded Options and Option-Adjusted Measures

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Learning Objectives

After completing this lecture you should be able to:

- Explain why embedded options make bond cash flows state-dependent and why today's discount curve is no longer enough.
- Explain why traditional yield-spread analysis fails when one yield is used to summarize cash flows at many dates.
- Interpret the **static spread** as the constant spread over the spot curve that equates promised cash flows to market price.
- Describe the investment characteristics of callable and putable bonds, including capped upside and negative convexity for callable bonds.

- Decompose embedded-option bonds into an option-free bond plus or minus the value of the embedded option.
- Use a short-rate lattice and backward induction to value option-free, callable, and putable bonds.
- Apply the node-level exercise rule: callable bonds use a minimum rule and putable bonds use a maximum rule.
- Explain why a valuation-grade interest-rate tree must be calibrated to the zero curve, not only generated from r_0, σ, and Δt.
- Interpret the **option-adjusted spread (OAS)** as the spread that reconciles model value with market price after option exercise is modeled.
- Explain why **effective duration** and **effective convexity** are repricing-based risk measures for bonds whose cash flows can change.

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1 Big Picture: Why Embedded Options Change Everything

In Lecture 7, the main idea was that a plain fixed-rate bond usually does **not** require a stochastic interest-rate model. If the cash flows are fixed, valuation is mostly discounted cash flow:

$$P = \sum_i CF_i P(0, t_i).$$

Bonds with embedded options are different. Their cash flows are not fully known today because someone has a future choice:

- the issuer may call the bond;
- the investor may put the bond back to the issuer;
- mortgage borrowers may prepay;
- a structured note may change coupons based on future rates.

The option makes the bond's cash flows **state-dependent**. We do not only need today's discount curve. We need possible future interest-rate states, future bond values, and an exercise rule.

> **Core Intuition**
>
> A plain bond asks: "What are the promised cash flows worth today?"
> A bond with an embedded option asks: "What might rates be in the future, what would the bond be worth in each future state, and would someone exercise the option?"

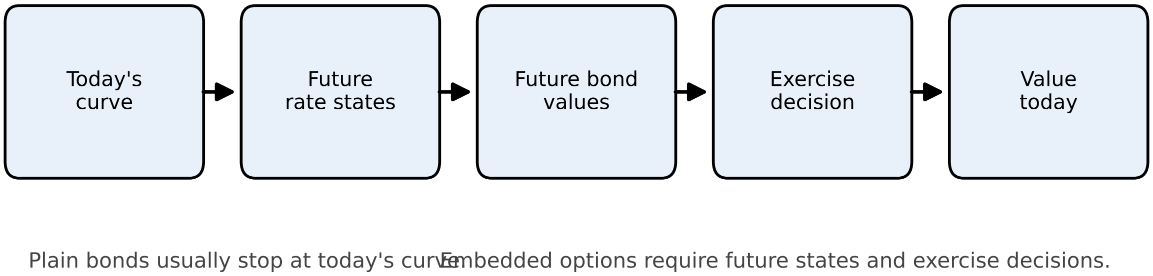

Figure 1: Why embedded options require a model of future rates

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2 Drawbacks of Traditional Yield-Spread Analysis

Historically, fixed-income analysts compared the **yield to maturity (YTM)** of a corporate bond with the yield of a Treasury security of similar maturity to measure compensation for credit and liquidity risk. The **yield spread** was defined as

$$\text{Yield spread} = y_{\text{corporate}} - y_{\text{Treasury}}$$

While straightforward, this measure suffers from several drawbacks:

1. **One-rate summary.** Yield to maturity compresses all of a bond's cash flows into a single average return. That single rate ignores that coupons and principal arrive at different dates and should be discounted using maturity-specific spot rates. It also embeds a reinvestment assumption: coupons are assumed to be reinvested at the YTM until maturity.
2. **Ignores embedded options.** Callable or putable bonds contain options that significantly affect cash flow timing and risk. Comparing YTMs of a callable corporate bond and a noncallable Treasury ignores the call option's value. The resulting yield spread blends credit, liquidity, and option premiums, making interpretation difficult.
3. **Assumes parallel shifts.** Yield spread analysis implicitly treats interest-rate movements too simply. In reality, yield curves twist, steepen, flatten, and move non-parallel across maturities.

4. **Fails to separate risk components.** Credit risk, liquidity risk, and option risk vary across issuers and over time. A simple yield spread bundles them together.

Because of these limitations, modern valuation methods use term-structure models and option-adjusted measures to isolate risks and provide more consistent valuations.

The practical problem is that a quoted yield spread is a **mixture**. For a callable corporate bond, the spread may compensate investors for credit risk, liquidity risk, and the fact that the investor is short the issuer's call option. A single YTM spread does not tell us how much comes from each source.

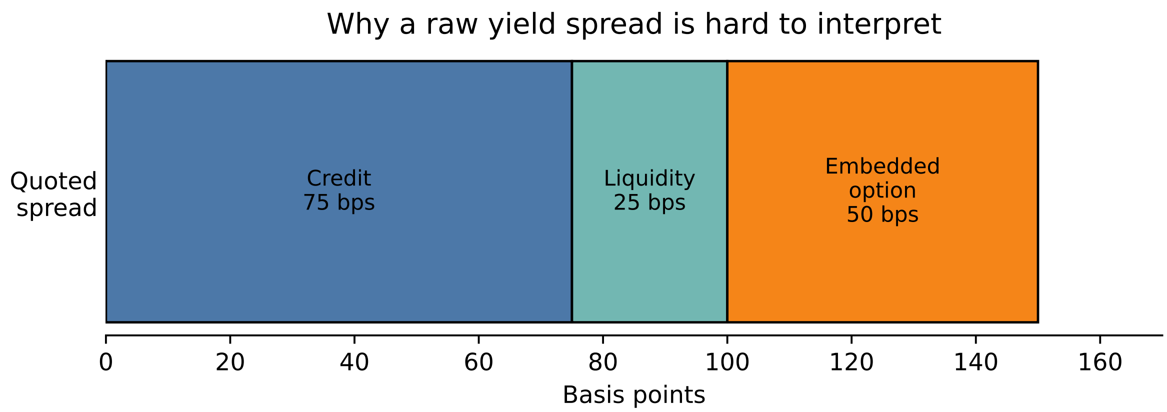

Figure 2: A quoted yield spread bundles together several sources of compensation

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3 Static Spread: An Alternative to Yield Spread

The **static spread** (also called the **zero-volatility spread**) measures the constant margin over the Treasury spot curve that discounts a bond's cash flows to its market price. For a bond with cash flows $\{CF_{t_i}\}$ at times $t_1, t_2, \ldots, t_n$ and Treasury zero rates $\{r_{t_i}\}$, the static spread s solves

$$P = \sum_{i=1}^{n} \frac{CF_{t_i}}{(1 + r_{t_i} + s)^{t_i}}$$

where P is the observed price.

Because the same spread s is added to all discount rates, the method accounts for the shape of the term structure while producing a single summary measure of compensation for non-Treasury risks. Unlike the yield spread, the static spread discounts each cash flow at a maturity-appropriate rate. That makes it more economically meaningful for coupon bonds, amortizing bonds, and bonds whose cash flows are distributed over time.

However, the static spread still has an important weakness: it discounts **promised** cash flows, not **expected** cash flows. For a callable bond, the promised maturity cash flow may never be received if the issuer refinances and calls the bond when rates fall.

Static Spread Intuition

Static spread improves on a simple yield spread because each cash flow is discounted using the spot rate for its own maturity. But it still treats the cash-flow schedule as fixed. That is why it works better for option-free bonds than for callable, putable, or mortgage-backed securities.

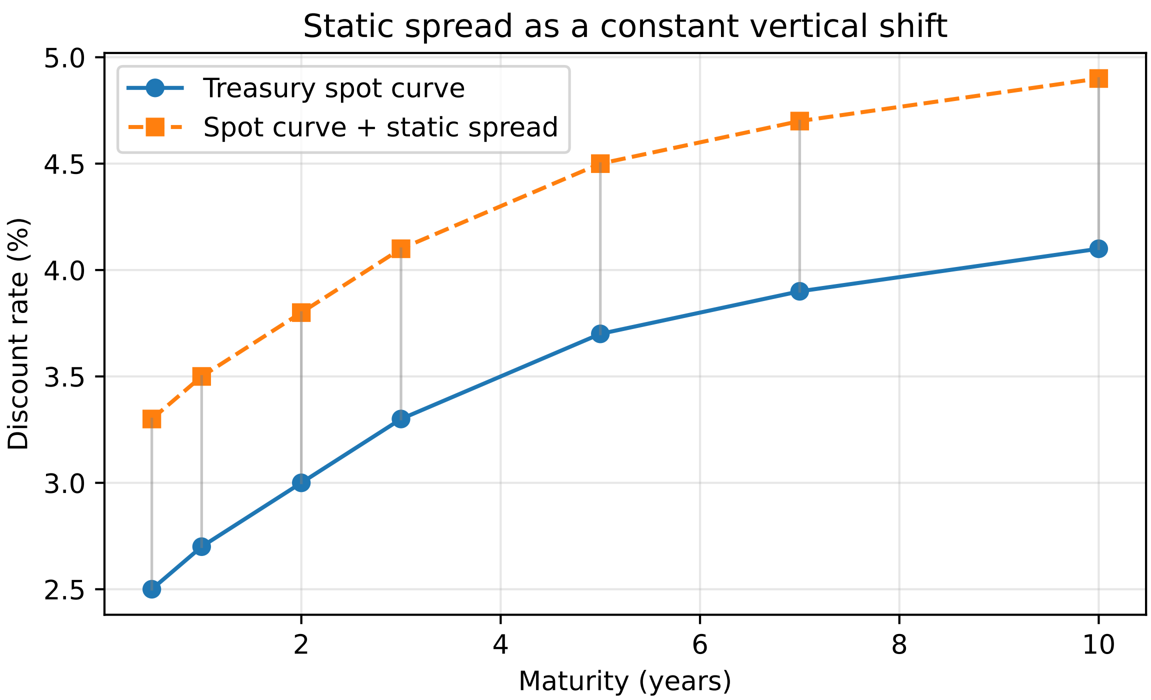

Figure 3: Static spread adds the same margin to each point on the spot curve

Example 1: Computing the static spread

Suppose a five-year corporate bond has semiannual coupons of 6 percent, so it pays \$30 every six months on a \$1,000 face value. The Treasury spot curve for maturities 0.5 years to 5 years is:

$$\{2.0\%, 2.2\%, 2.6\%, 3.0\%, 3.4\%, 3.8\%, 4.0\%, 4.1\%, 4.2\%, 4.3\%\}$$

Assume the bond trades at \$1,020. The static spread is the single number s that makes the present value of the promised cash flows equal to the observed market price:

$$1020 = \sum_{i=1}^{10} \frac{CF_i}{(1 + r_i + s)^{t_i}}.$$

The graph below plots the model price for different trial spreads. The solid blue line is the present value of the bond's promised cash flows as the assumed spread changes. This relationship is **not linear**:

$$PV(s) = \sum_{i=1}^{10} \frac{CF_i}{(1 + r_i + s)^{t_i}}.$$

It may look nearly linear over a narrow spread range because a smooth curve is well approximated by a tangent line locally. The horizontal dashed line is the observed market price. The vertical dashed line marks the spread where the model price intersects the observed price.

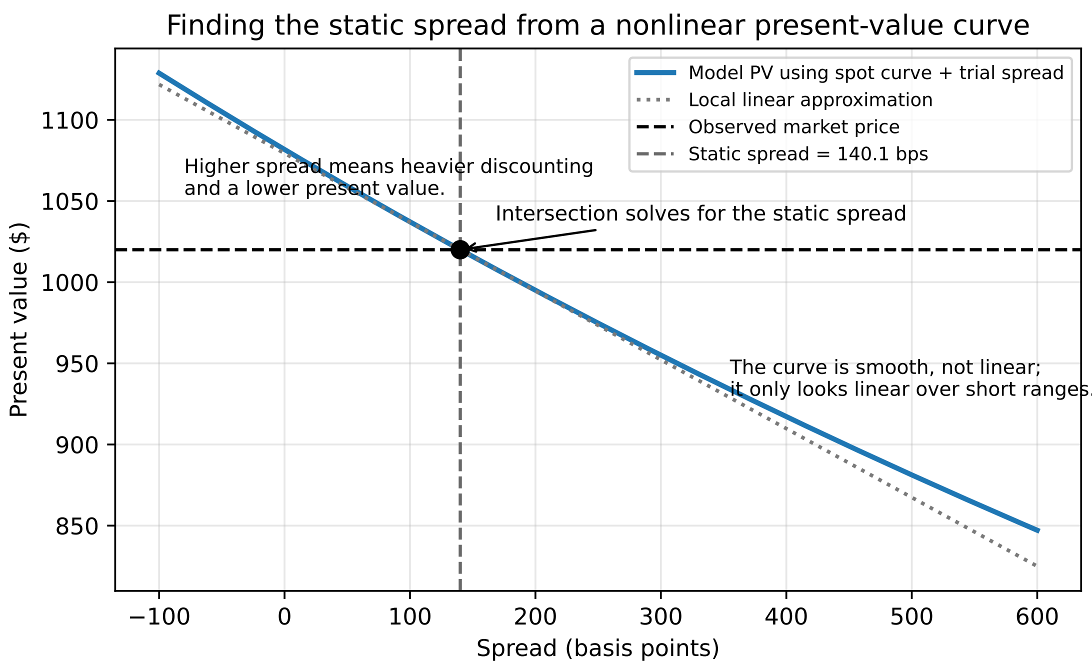

Figure 4: Static spread is the spread where model present value equals the observed price

In this example, the static spread is about 140 basis points. Economically, investors require roughly 140 basis points over the Treasury spot curve to hold this bond at a price of $1,020.

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4 Callable Bonds and Their Investment Characteristics

A **callable bond** gives the issuer the right, but not the obligation, to redeem the bond before maturity at a specified call price according to a call schedule. The issuer will typically call the bond when prevailing rates fall below the coupon rate, allowing refinancing at a lower borrowing cost.

This embedded call option changes the bond's economic behavior.

> **Borrower and Lender Intuition**
>
> The issuer is like a homeowner with the right to refinance a mortgage. If rates fall, refinancing is attractive to the borrower. But that same event hurts the lender, because the lender loses a high-coupon asset and must reinvest at lower rates.

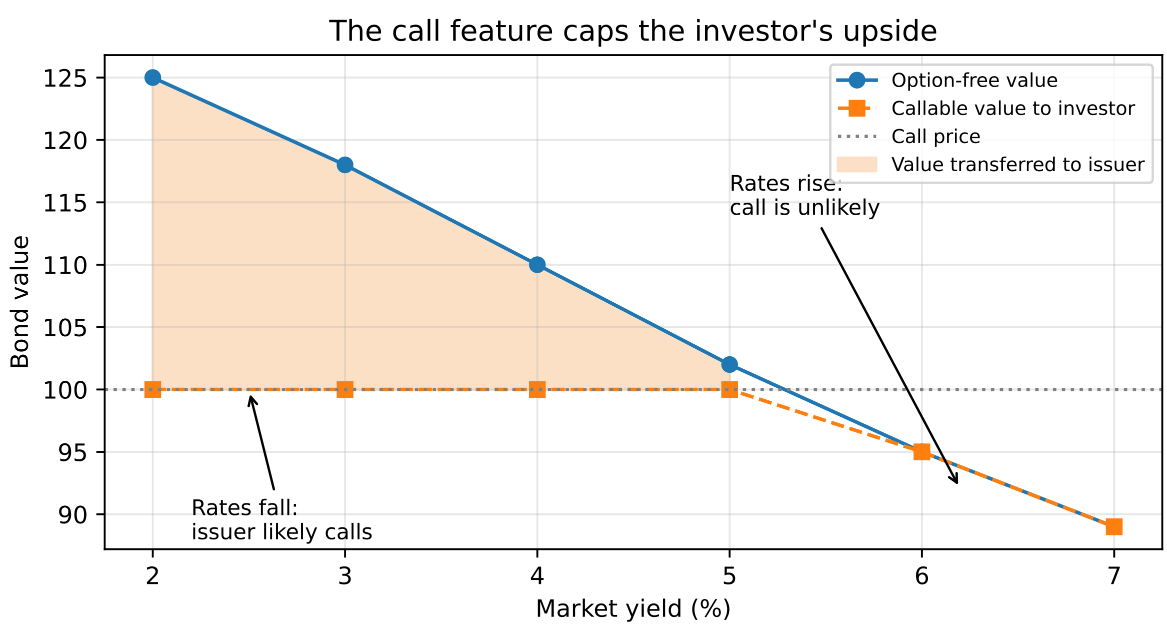

Figure 5: Callable bond economics: falling rates help the issuer and limit investor upside

4.1 Lower price than a comparable option-free bond

Because the issuer owns the right to call, the investor is effectively **short a call option**. The callable bond therefore must be worth less than an otherwise identical option-free bond:

$$V_{\text{callable}} = V_{\text{option-free}} - V_{\text{call option}}$$

4.2 Negative convexity

As yields decline, the price of an option-free bond rises at an increasing rate. A callable bond does not. Once rates fall enough, the chance of being called rises sharply, which caps further price appreciation. This gives callable bonds **negative convexity** over some yield ranges.

4.3 Yield to worst

Because a callable bond may be redeemed at one of several call dates, investors often compute:

- yield to maturity
- yield to each call date
- **yield to worst**, which is the lowest among these yields

Yield to worst is a practical convention, but it is not a full valuation framework. It does not explicitly model volatility or the contingent nature of cash flows.

4.4 Price-yield relationship for a callable bond

To visualize negative convexity, we can compare the price-yield curve of an option-free bond with that of a callable bond.

The callable bond's price is capped when yields fall enough for calling to become attractive to the issuer. This is the key intuition behind negative convexity.

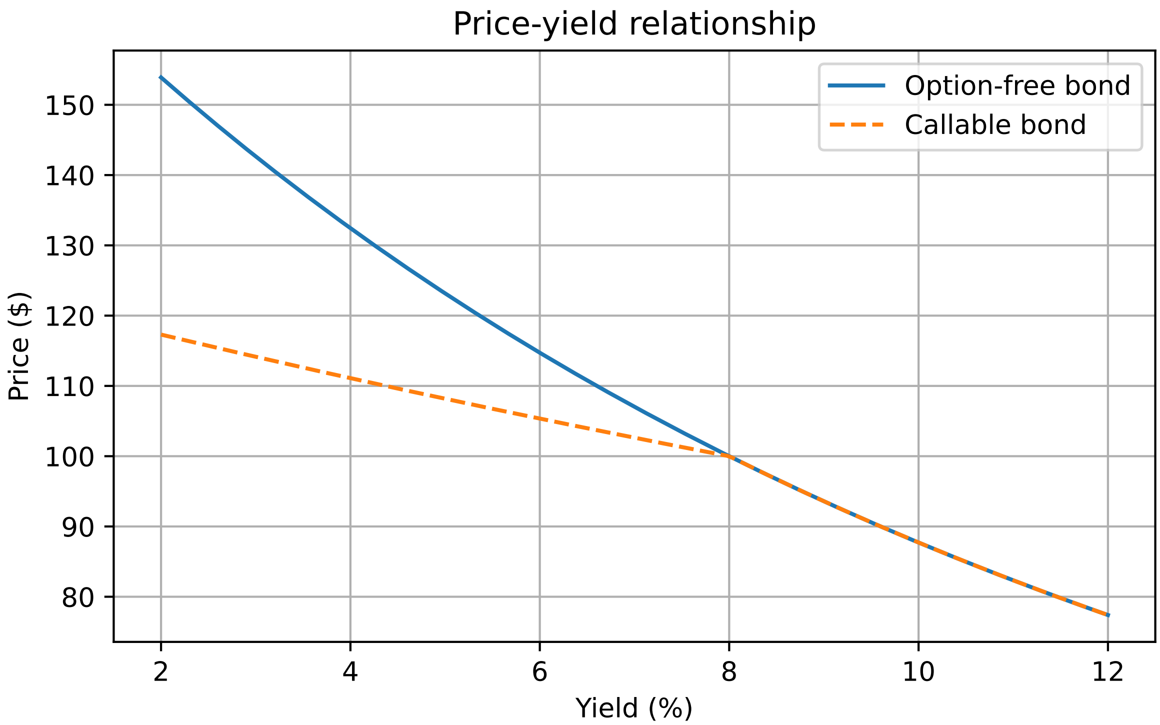

Figure 6: Price-yield relationship for an option-free bond and a callable bond

5 Components of a Bond with an Embedded Option

A bond with an embedded option can be decomposed conceptually into two parts:

1. **Option-free bond.** This is the value of the bond if no option existed.
2. **Embedded option.** This is the value of the call or put written into the contract.

For a callable bond:

$$V_{\text{callable}} = V_{\text{straight}} - V_{\text{call option}}$$

For a putable bond:

$$V_{\text{putable}} = V_{\text{straight}} + V_{\text{put option}}$$

This decomposition is useful because it tells us exactly where the pricing distortion comes from. The bond itself generates fixed-income cash flows. The option changes the timing and certainty of those cash flows.

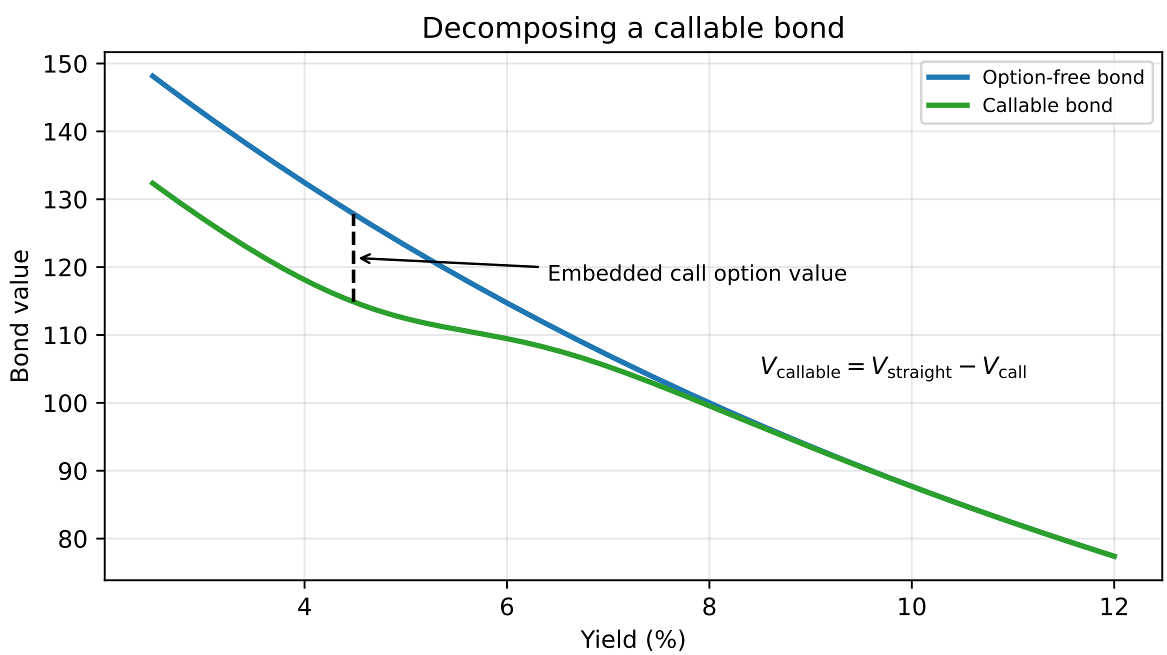

Figure 7: For a callable bond, the embedded call option is the gap between straight-bond value and callable-bond value

6 Valuation Model for Bonds with Embedded Options

The modern approach to valuing bonds with embedded options uses an **interest-rate lattice** and **risk-neutral valuation**.

The general logic is:

1. Build a tree of future short rates.
2. Determine future bond values at the terminal nodes.
3. Move backward through the tree using risk-neutral expectations.
4. At each node, apply the embedded option rule:
   - callable bond: issuer chooses the lower of holding value and call price
   - putable bond: investor chooses the higher of holding value and put price

The value at time 0 is the bond's model price.

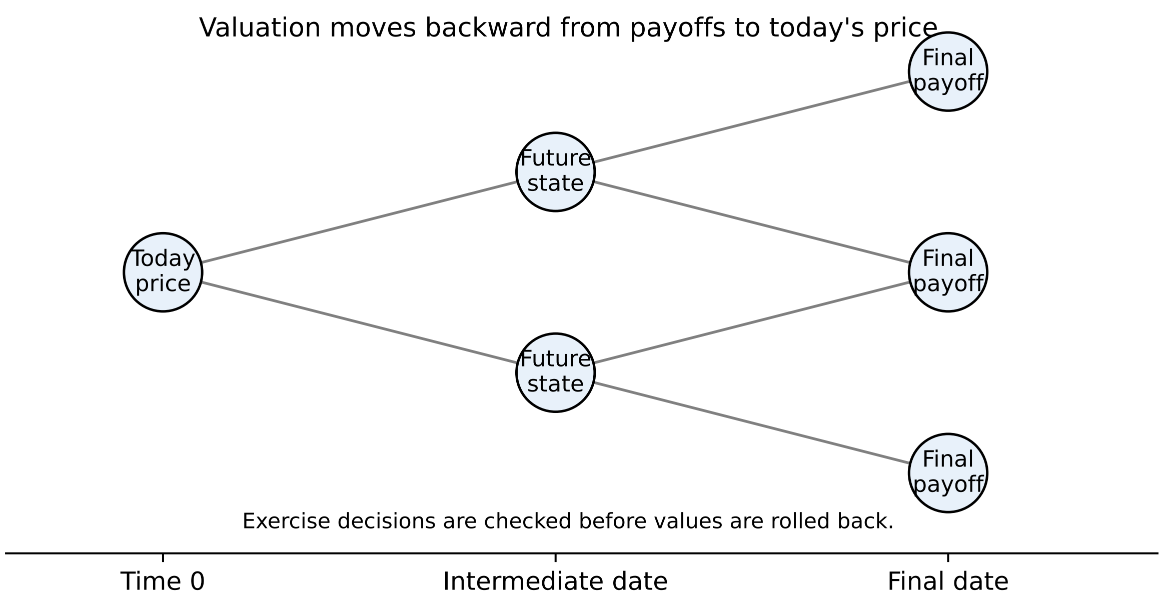

Figure 8: Backward induction: start with future payoffs, then work back to today

6.1 The key backward induction formula

For an option-free bond, backward induction uses

$$V_{i,j} = \frac{p\,V_{i+1,j+1} + (1-p)\,V_{i+1,j}}{1+r_{i,j}} + CF_i$$

where:

- $V_{i,j}$ is the bond value at time step i and state j
- p is the risk-neutral probability of an up move
- $r_{i,j}$ is the short rate at node (i,j)
- CF_i is the coupon paid at that date

For a callable bond:

$$V_{i,j}^{\text{callable}} = \min\left\{\frac{p\,V_{i+1,j+1} + (1-p)\,V_{i+1,j}}{1+r_{i,j}} + CF_i,\ \text{Call Price}\right\}$$

For a putable bond:

$$V_{i,j}^{\text{putable}} = \max\left\{\frac{p\,V_{i+1,j+1} + (1-p)\,V_{i+1,j}}{1+r_{i,j}} + CF_i,\ \text{Put Price}\right\}$$

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7 Why the lattice matters

The lattice is not just a computational trick. It is economically important.

At each date, interest rates may move up or down. That means the bond's future value is not fixed today. The option's value comes from this uncertainty.

The lattice lets us model three things at the same time:

- the time value of money
- uncertainty in future short rates
- optimal exercise of the embedded option

This is why spread measures based only on yield to maturity are inadequate for callable and putable bonds.

A useful way to think about the tree is this:

- the horizontal direction represents **time**
- the vertical direction represents **interest-rate states**

- each node represents a possible short rate that could prevail at that future date
- at each node, someone makes an economic decision:
  - the issuer decides whether to call
  - the investor decides whether to put
  - if neither side exercises, the bond continues

So the embedded option makes the bond **path-dependent**. The value depends not only on maturity and coupon, but also on what happens to rates along the way.

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8 Constructing a simple binomial interest-rate tree

To illustrate, consider a three-year bond with annual coupons of 8 on a face value of 100. Assume annual time steps. We first construct a short-rate tree directly for intuition.

Suppose the short-rate lattice is:

State	Year 0	Year 1	Year 2	Year 3
Low-rate state	6%	5%	4%	3%
Middle state		9%	8%	7%
High-rate state			12%	11%
Very high-rate state				15%

Assume:

- face value = 100
- annual coupon = 8
- callable at 107
- first callable date = year 2
- risk-neutral probability $p = 0.5$

This table puts time horizontally and states vertically. At each date, the node value is computed from the two adjacent nodes one year ahead.

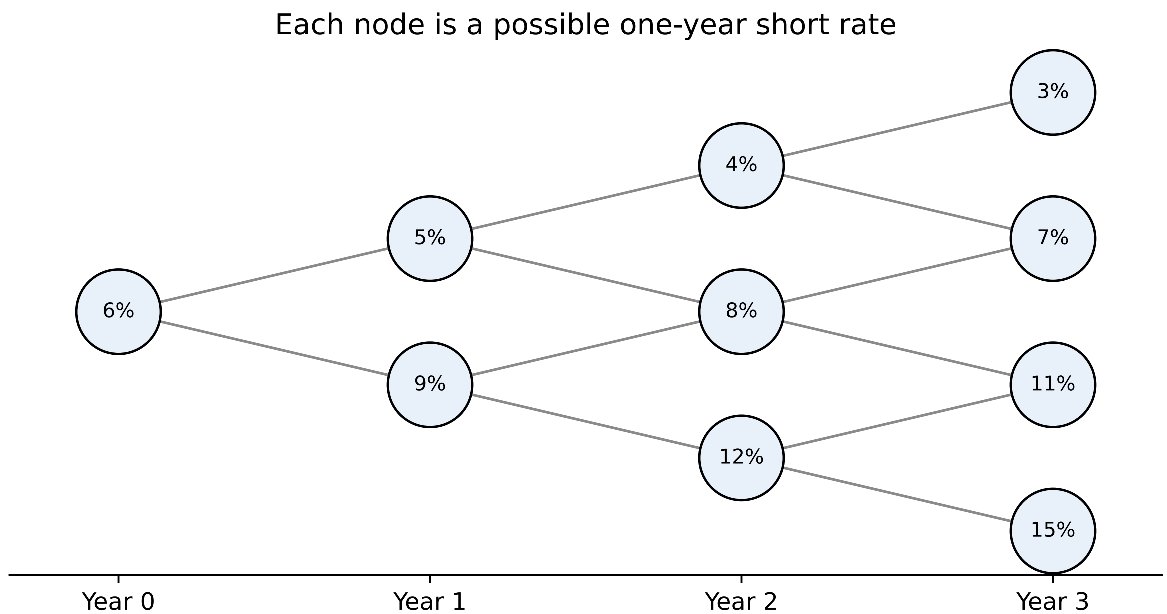


Figure 9: Short-rate lattice used in the worked example

9 Step-by-step numerical example: pricing an option-free bond and a callable bond

We now price the bond by hand using backward induction. The point of the example is to see exactly where the option changes the cash flows.

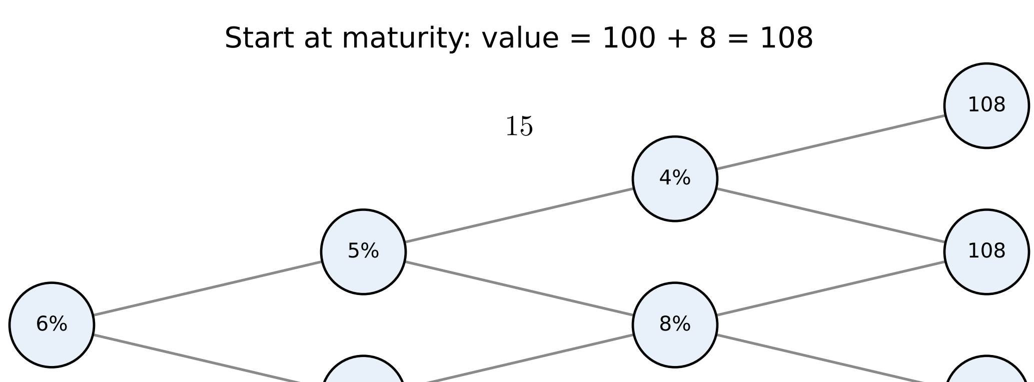
9.1 Step 1: Terminal values at maturity

At year 3 the bond pays final coupon plus principal:

$$108 = 100 + 8$$

So every year-3 terminal node has the same payoff:

State	Year-3 payoff
Low-rate state	108.00
Middle state	108.00
High-rate state	108.00
Very high-rate state	108.00



9.2 Step 2: Move backward to year 2 for the option-free bond

At year 2, the option-free hold value is:

$$V = \frac{0.5 \times 108 + 0.5 \times 108}{1 + r} + 8 = \frac{108}{1 + r} + 8$$

Year 2, rate = 4%

$$V = \frac{108}{1.04} + 8 = 103.8462 + 8 = 111.8462$$

Year 2, rate = 8%

$$V = \frac{108}{1.08} + 8 = 100.0000 + 8 = 108.0000$$

Year 2, rate = 12%

$$V = \frac{108}{1.12} + 8 = 96.4286 + 8 = 104.4286$$

So the option-free values at year 2 are:

Node at Year 2	Hold Value
4% state	111.85
8% state	108.00
12% state	104.43

9.3 Step 3: Apply the call feature at year 2

Because the bond is callable at 107 from year 2 onward, the issuer compares the hold value with the call price. The issuer calls only if calling is cheaper than leaving the bond outstanding.

For the callable bond:

$$V^{\text{callable}} = \min\{\text{Hold Value},\ 107\}$$

Thus:

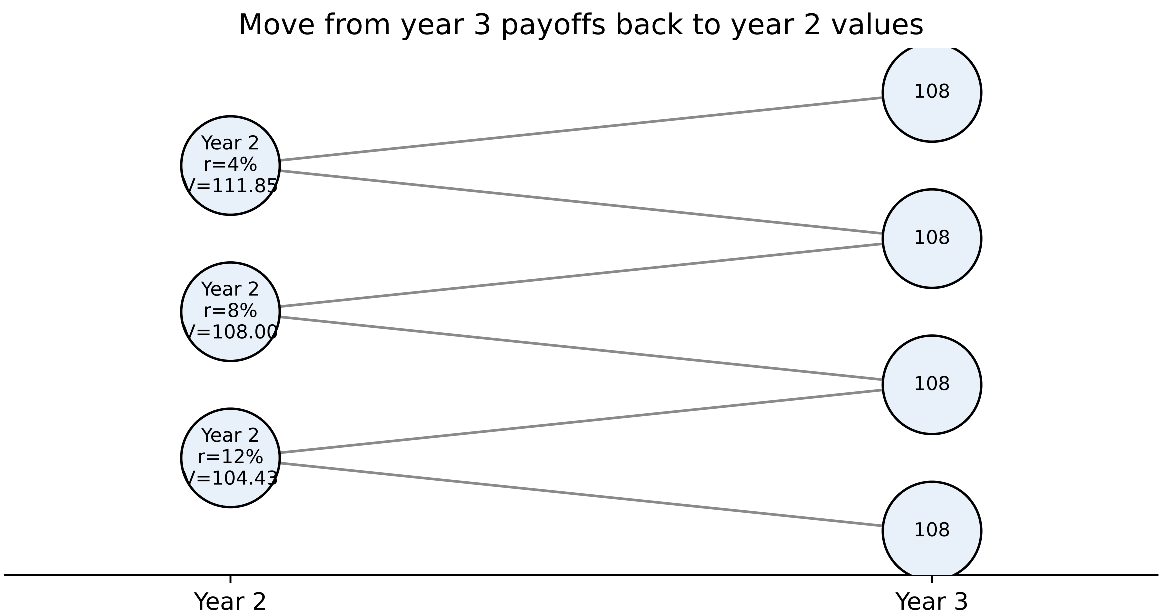


Figure 11: Step 2: each year-2 value is computed from the two adjacent terminal nodes

Node at Year 2	Hold Value	Callable Value
4% state	111.85	107.00
8% state	108.00	107.00
12% state	104.43	104.43

This is the key economic point: the issuer calls in the low-rate states but not in the high-rate state. The option exercise decision is state-dependent.

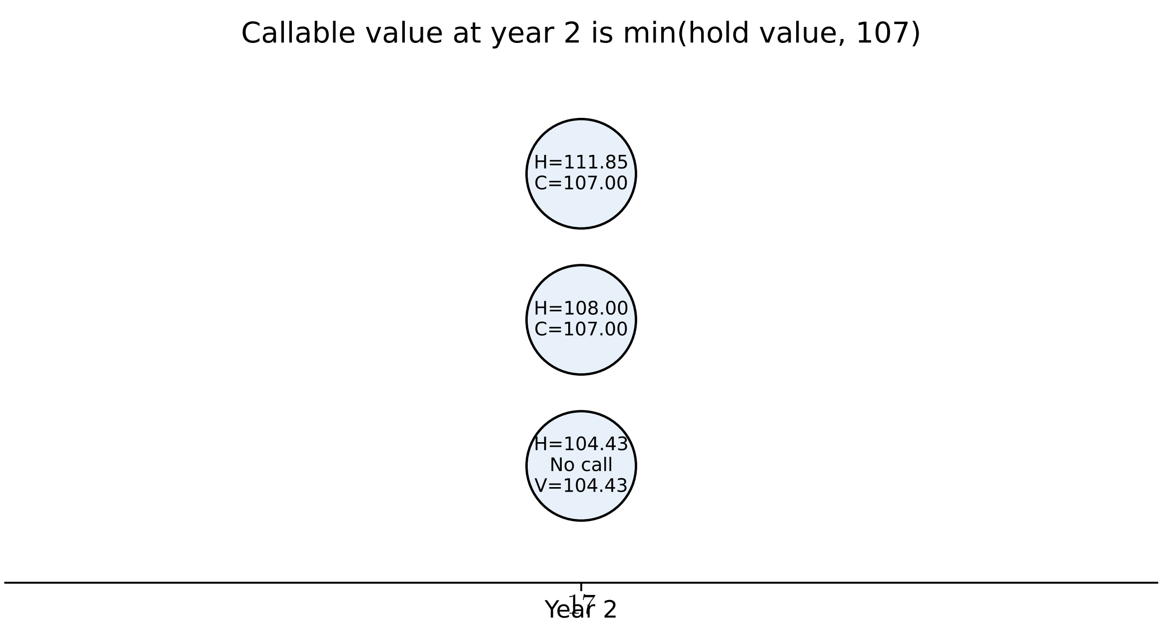


Figure 12: Step 3: the call rule caps year-2 values above the call price

9.4 Step 4: Move backward to year 1 for the option-free bond

Now compute the option-free bond values at year 1 using the option-free year-2 values.

Year 1, higher-rate node, $r = 9\%$

$$V = \frac{0.5(104.4286) + 0.5(108.0000)}{1.09} + 8$$

$$V = \frac{106.2143}{1.09} + 8 = 97.4443 + 8 = 105.4443$$

So:

Node at Year 1	Option-Free Value
5% state	112.69
9% state	105.44

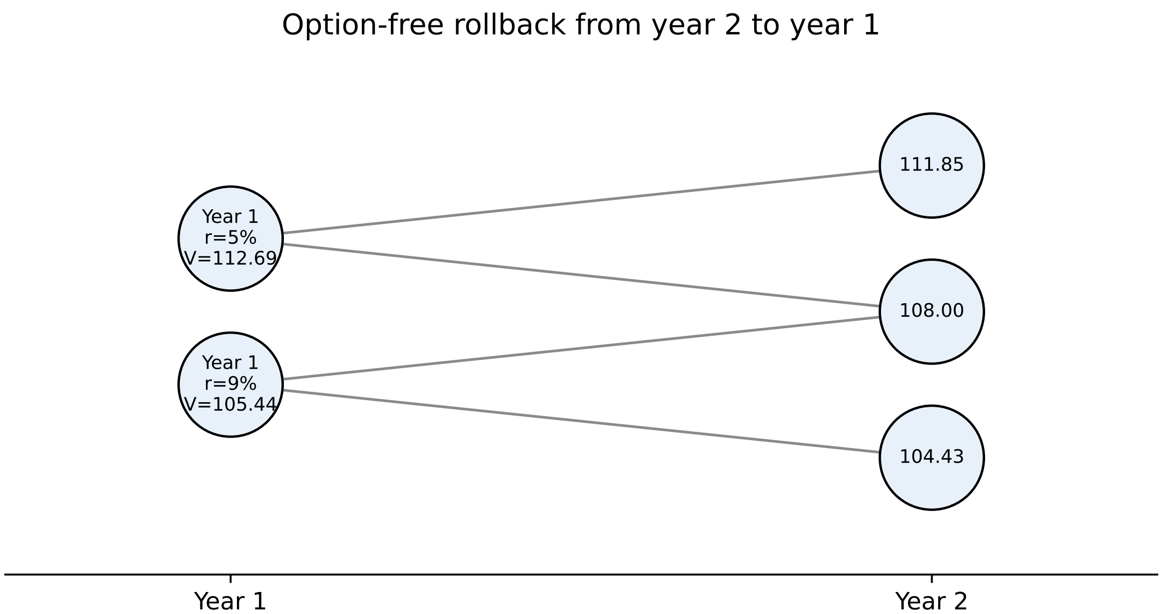


Figure 13: Step 4: option-free year-1 values are rolled back from option-free year-2 values

9.5 Step 5: Move backward to year 1 for the callable bond

Now do the same using the callable values at year 2.

Year 1, rate = 5%

$$V = \frac{0.5(107.0000) + 0.5(107.0000)}{1.05} + 8 = \frac{107}{1.05} + 8 = 101.9048 + 8 = 109.9048$$

Year 1, rate = 9%

$$V = \frac{0.5(104.4286) + 0.5(107.0000)}{1.09} + 8 = \frac{105.7143}{1.09} + 8 = 96.9856 + 8 = 104.9856$$

So:

Node at Year 1	Callable Value
5% state	109.90
9% state	104.99

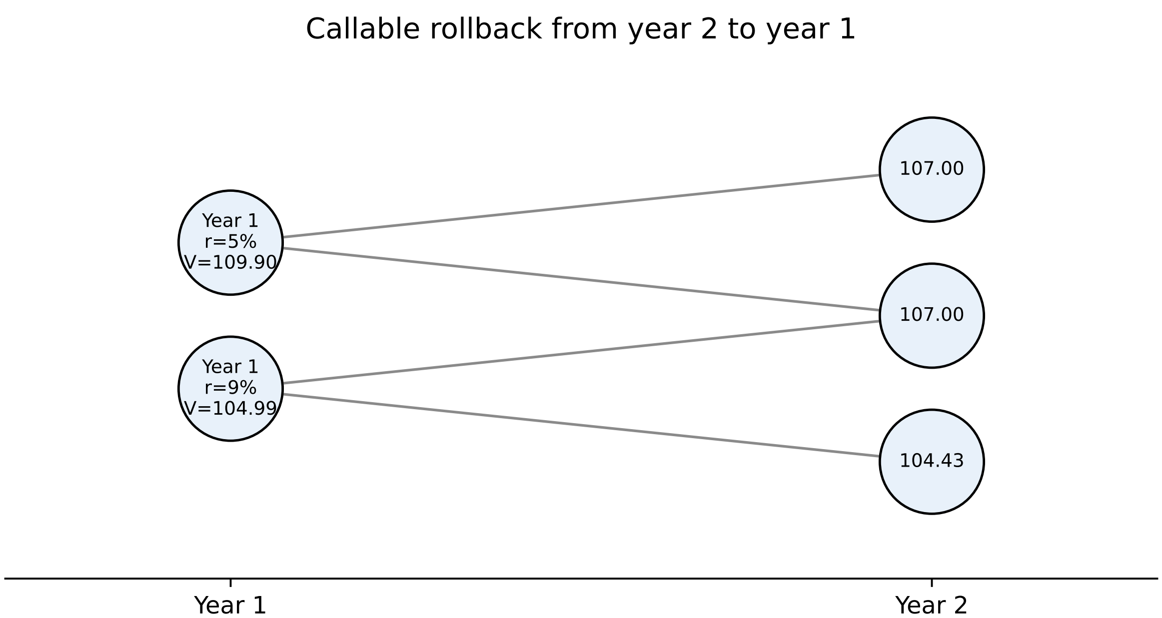


Figure 14: Step 5: callable year-1 values use the call-adjusted year-2 values

9.6 Step 6: Move backward to time 0

Option-free bond at time 0

$$V_0^{\text{straight}} = \frac{0.5(112.6886) + 0.5(105.4443)}{1.06} + 8$$

$$V_0^{\text{straight}} = \frac{109.0665}{1.06} + 8 = 102.8930 + 8 = 110.8930$$

Callable bond at time 0

$$V_0^{\text{callable}} = \frac{0.5(109.9048) + 0.5(104.9856)}{1.06} + 8$$

$$V_0^{\text{callable}} = \frac{107.4452}{1.06} + 8 = 101.3634 + 8 = 109.3634$$

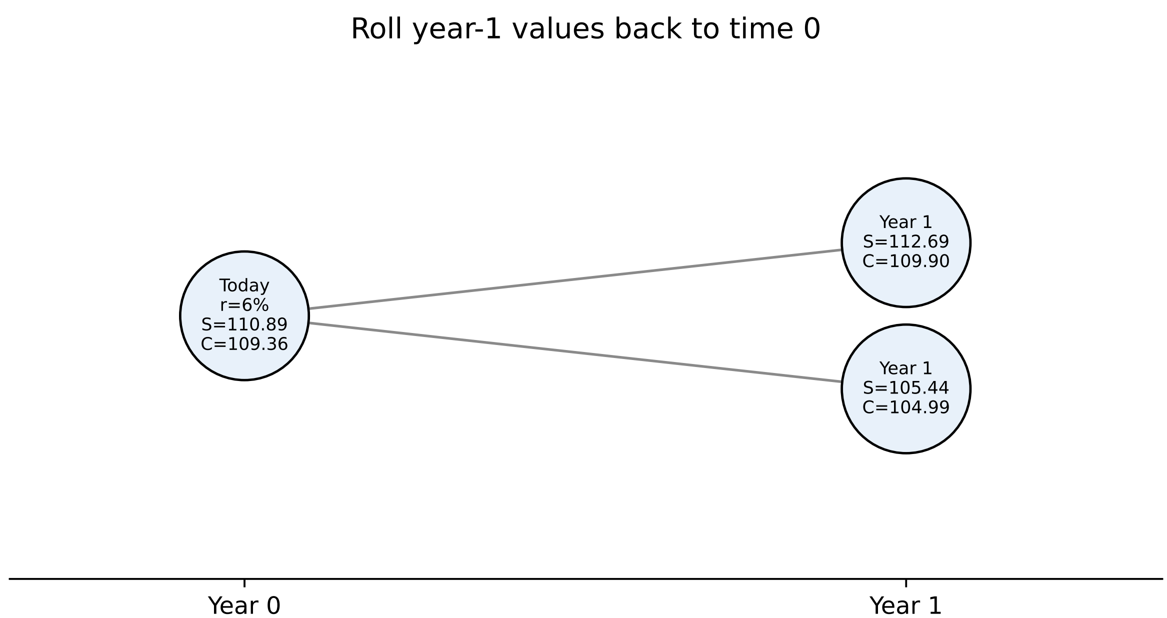


Figure 15: Step 6: final rollback gives today's option-free and callable values

9.7 Step 7: Embedded call option value

The call option value is the difference between the straight bond value and the callable bond value:

$$V_{\text{call option}} = V_{\text{straight}} - V_{\text{callable}}$$

$$V_{\text{call option}} = 110.8930 - 109.3634 = 1.5296$$

So in this example:

Quantity	Value
Option-free bond price	110.89
Callable bond price	109.36
Embedded call option value	1.53

! Interpretation of the numerical example

- The callable bond is worth less than the straight bond because the issuer owns the call option.
- The option matters most in low-rate states. At year 2, the issuer calls in the 4% and 8% states because the hold value is above the call price of 107. At 12%, the issuer does not call because the bond is worth only 104.43 if left outstanding.
- The lattice makes the exercise decision explicit at each node. That is the real advantage of the method: the model does not assume one call date mechanically; it checks whether exercise is economically optimal in each state.

i Worked Example: Pricing a Putable Bond on a Lattice

Now consider a two-year annual-pay bond with:

- face value = 100
- annual coupon = 5
- current one-year short rate = 6%
- year-1 short rates: 4% in the low-rate state and 12% in the high-rate state
- risk-neutral probability $p = 0.5$
- putable at 100 in year 1

The short-rate lattice is:

State	Year 0	Year 1	Year 2
Low-rate state	6%	4%	
High-rate state		12%	

At maturity, the payoff is:

$$105 = 100 + 5.$$

At year 1, first compute the continuation value before applying the put:

$$V_{\text{hold}} = \frac{105}{1+r} + 5.$$

For the low-rate node:

$$V_{\text{hold, low}} = \frac{105}{1.04} + 5 = 105.9615.$$

For the high-rate node:

$$V_{\text{hold, high}} = \frac{105}{1.12} + 5 = 98.7500.$$

The investor keeps the bond in the low-rate state, but puts the bond back to the issuer in the high-rate state:

Year-1 state	Hold value	Put value	Putable value
4% state	105.96	100.00	105.96
12% state	98.75	100.00	100.00

Finally, roll the year-1 values back to today:

$$V_0^{\text{putable}} = \frac{0.5(105.9615) + 0.5(100.0000)}{1.06} + 5 = 102.1517.$$

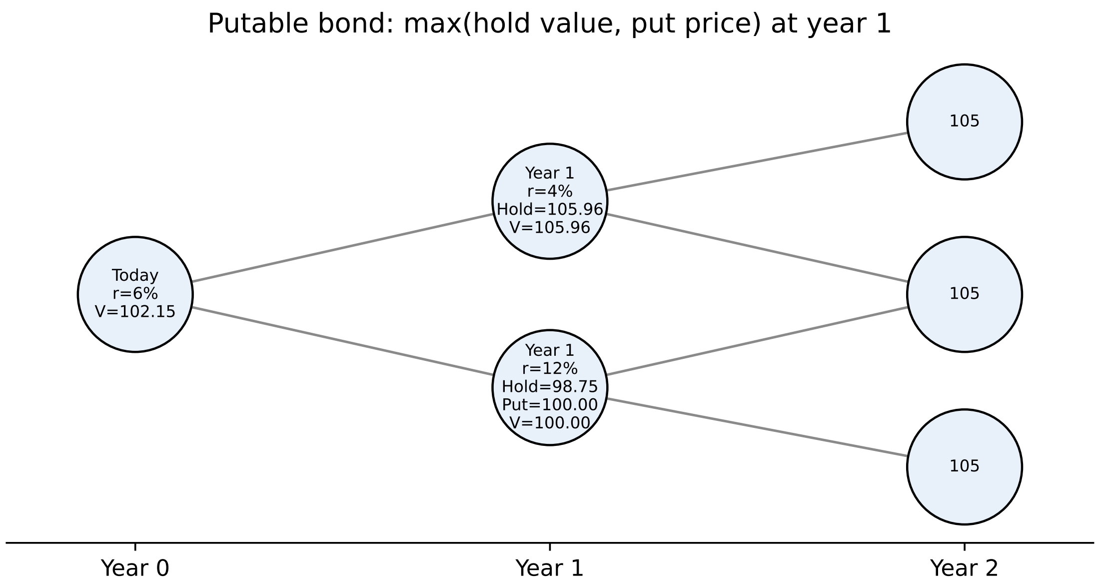


Figure 16: Worked Example 2: pricing a putable bond by backward induction

This second example shows that the same lattice logic works for an investor-owned option. A callable bond uses a **minimum** rule because the issuer chooses the lower cost. A putable bond uses a **maximum** rule because the investor chooses the higher value.

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10 Constructing a model-based binomial tree

The previous tree was chosen for teaching. A model-based tree is different: it is built so that it is consistent with observable market prices.

The inputs are:

- today's benchmark spot curve or issuer curve;
- a short-rate model;
- an assumed interest-rate volatility;

- a time step, such as one year, six months, or one month.

The first node is today's one-period short rate. From that node, the model allows rates to move up or down. A common lognormal specification uses an up factor and a down factor:

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \frac{1}{u}$$

where σ is the assumed volatility and Δt is the length of one time step. Higher volatility spreads the upper and lower nodes farther apart. Lower volatility keeps the tree tighter.

Calibration then chooses the level of rates at each future date so that the tree prices the market's on-the-run issues correctly. This is why the tree is called **arbitrage-free**: if the model could not reproduce the prices of the benchmark bonds used to build it, the model would be internally inconsistent with the observed curve.

The construction is iterative:

1. Start with today's one-period rate.
2. Choose a trial value for the lower future short rate.
3. Use the volatility assumption to determine the corresponding higher rate.
4. Price the next benchmark bond by backward induction.
5. Adjust the trial rate until the model price equals the observed market price.
6. Repeat the process one maturity at a time to extend the tree.

For example, suppose one-year rates can move up or down after one year. If the lower one-year forward rate one year from now is $r_{1,L}$, a lognormal tree with annual volatility σ sets the upper rate as:

$$r_{1,H} = r_{1,L}e^{2\sigma}.$$

The factor $e^{2\sigma}$ creates the spread between the low and high future rates. If $\sigma = 10\%$ and $r_{1,L} = 4.074\%$, then:

$$r_{1,H} = 4.074\% \times e^{0.20} = 4.976\%.$$

This pair of rates is not chosen because it looks plausible. It is accepted only if, when used in the tree, it correctly prices the relevant two-year benchmark bond. The same logic is then used to find the next set of rates for year 2, and so on.

The key intuition is that the tree has two jobs. It must represent possible future rate uncertainty, and it must remain anchored to today's observed term structure. A tree that only simulates rates but does not fit current bond prices is not suitable for valuation.

Example: Calibrating Four Maturities of the Zero Curve

Suppose the annual zero curve is:

Maturity	Zero rate	Market discount factor
1 year	4.00%	0.96154
2 years	4.50%	0.91573
3 years	5.00%	0.86384
4 years	5.25%	0.81469

A tree with only r_0, σ, and Δt cannot generally fit all four discount factors. A calibrated tree adds one level parameter per time step. A simple way to write this is:

$$r_{t,j} = a_t e^{(2j-t)\sigma},$$

where a_t is the level parameter for year t, j indexes the state, and σ controls the spacing between states. The volatility fixes the relative distance between rates at the same date; calibration chooses a_t so the tree matches the market zero price for that maturity.
With annual steps, $\sigma = 10\%$, and equal risk-neutral probabilities, one calibrated set of short rates is:

	Maturity	Market DF	Calibrated level a_t	Short rates at that date	Tree DF
0	1 year	0.96154	4.00%	4.00%	0.96154
1	2 years	0.91573	4.98%	4.51%, 5.50%	0.91573
2	3 years	0.86384	5.96%	4.88%, 5.96%, 7.28%	0.86384
3	4 years	0.81491	5.93%	4.40%, 5.37%, 6.56%, 8.01%	0.81491

The important point is not the specific numbers. The important point is the number of calibration knobs. With four maturities, the tree uses four level parameters, a_0, a_1, a_2, a_3, to fit four market discount factors. The volatility parameter still matters, but it controls the spread between states, not the overall level needed to match the zero curve.

In practice, professional valuation systems use calibrated models such as Ho-Lee, Black-Derman-Toy, Hull-White, or Black-Karasinski. The teaching objective here is not the full calibration machinery, but the economic structure: volatility determines the spacing of nodes, while calibration determines the level of the tree.

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11 Incorporating default risk

The lattice framework can be extended to corporate bonds in several ways.

One approach is to add a credit spread s to each node's short rate:

$$r_{i,j}^{\text{risky}} = r_{i,j} + s$$

Another is to introduce default probabilities and recovery assumptions directly into the tree. Then the node value reflects both survival and default scenarios.

These extensions are important in practice because many callable bonds are corporate or structured products whose spreads reflect both credit and optionality.

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12 Implementation challenges

The classroom tree is intentionally small. The main implementation issues in practice are narrower and more specific:

- **Curve fit.** The model must reproduce the current benchmark curve or issuer curve used for valuation. A poor curve fit contaminates every node value.
- **Volatility input.** Callable-bond values are sensitive to assumed rate volatility. Higher volatility increases the value of the issuer's call option and lowers the callable bond's value.
- **Exercise rule.** The model must reflect the actual call schedule, call prices, notice periods, make-whole terms, and any restrictions on exercise.
- **Timing.** Real cash flows may not fall exactly on lattice dates. The model must handle coupon dates, call dates, and settlement dates without distorting discounting.
- **Credit treatment.** For corporate bonds, the analyst must decide whether credit risk enters through an issuer curve, an added spread, explicit default probabilities, or some combination.

These choices matter because OAS and effective duration are model outputs. They are only as reliable as the curve, volatility, and exercise assumptions used to produce them.

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13 Option-Adjusted Spread (OAS)

The **option-adjusted spread (OAS)** is the constant spread added to every short rate in the calibrated tree so that the model value equals the market price.

The word **option-adjusted** matters. For a callable bond, the model first allows future cash flows to change when the issuer optimally calls the bond. The spread is then solved after those option-dependent cash flows have been modeled.

Conceptually, OAS solves:

$$P_{\text{market}} = \text{model value using } (r_{i,j} + s_{\text{OAS}}) \text{ and the embedded-option exercise rule.}$$

This is different from a static spread. Static spread discounts the promised cash-flow schedule. OAS discounts model-implied cash flows that may change across interest-rate paths.

13.1 Why OAS is useful

Traditional yield spread has two major weaknesses for callable bonds. First, it compares one yield with one benchmark maturity, so it does not fully use the spot curve. Second, it ignores the fact that future cash flows may change when rates move. Static spread fixes the first problem but not the second. OAS is designed to address both.

For a callable bond, the quoted spread includes compensation for:

- credit and liquidity risk;
- uncertainty about future rates;
- the investor's short position in the issuer's call option.

OAS attempts to express the compensation left after the model has accounted for the embedded option.

13.2 Benchmark choice

The interpretation of OAS depends on the curve used to build the tree:

Tree built from	What OAS mainly reflects
Treasury or risk-free curve	Credit, liquidity, relative value, and any model error after option effects
Issuer curve	Richness or cheapness relative to that issuer's own curve

This is why an OAS quote is incomplete unless we know the benchmark curve, volatility assumption, and model.

13.3 Solving for OAS

The calculation is a root-finding problem:

1. Build an arbitrage-free short-rate tree from the chosen curve and volatility.
2. Add a trial spread to every node rate.
3. Value the bond by backward induction, applying the call or put rule at each exercise node.
4. Compare the model value with the market price.
5. Raise or lower the trial spread until the model value equals the market price.

13.4 Option cost in spread terms

For an option-free bond, static spread and OAS are usually close if the same curve is used. For a callable bond, the static spread is usually larger because it values the promised maturity cash flows and ignores that the issuer may call the bond.

The option cost in spread terms is often summarized as:

$$\text{Option cost} = \text{Static spread} - \text{OAS}.$$

For example, if a callable bond has a static spread of 150 basis points and an OAS of 100 basis points, then about 50 basis points compensate the investor for the embedded call option.

> **OAS Intuition**
>
> OAS is not the total spread investors see in a yield quote. It is the spread left after the model has allowed interest rates to move and has allowed the embedded option to change the bond's cash flows.

Higher assumed volatility generally lowers the OAS of a callable bond because the issuer's call option becomes more valuable. For a putable bond, higher volatility tends to raise the value of the investor's put option.

14 Effective Duration and Convexity

For bonds with embedded options, traditional Macaulay duration and convexity are inadequate because cash flows depend on the future path of interest rates.

Instead, we use **effective duration** and **effective convexity**, which are based on revaluing the bond after small changes in the entire yield curve.

Let:

- P_0 = base price
- P_- = price when the curve shifts **down** by Δy
- P_+ = price when the curve shifts **up** by Δy

Then:

$$D_e = \frac{P_- - P_+}{2P_0 \Delta y}$$

$$C_e = \frac{P_- + P_+ - 2P_0}{P_0 (\Delta y)^2}$$

For callable bonds, effective duration is often shorter than for an otherwise similar option-free bond, and effective convexity may become small or even negative when the bond is likely to be called.

> **i** Interactive Example: Four-Period Lattice Valuation
>
> The maturity is fixed at four annual periods. Change the bond and lattice assumptions to see how the short-rate tree, model price, OAS, effective duration, and effective convexity respond.
> This example is interactive in the HTML version of the lecture.

Because the call feature limits upside when rates fall, the callable bond's price response is asymmetric. That is exactly what effective convexity is designed to capture.

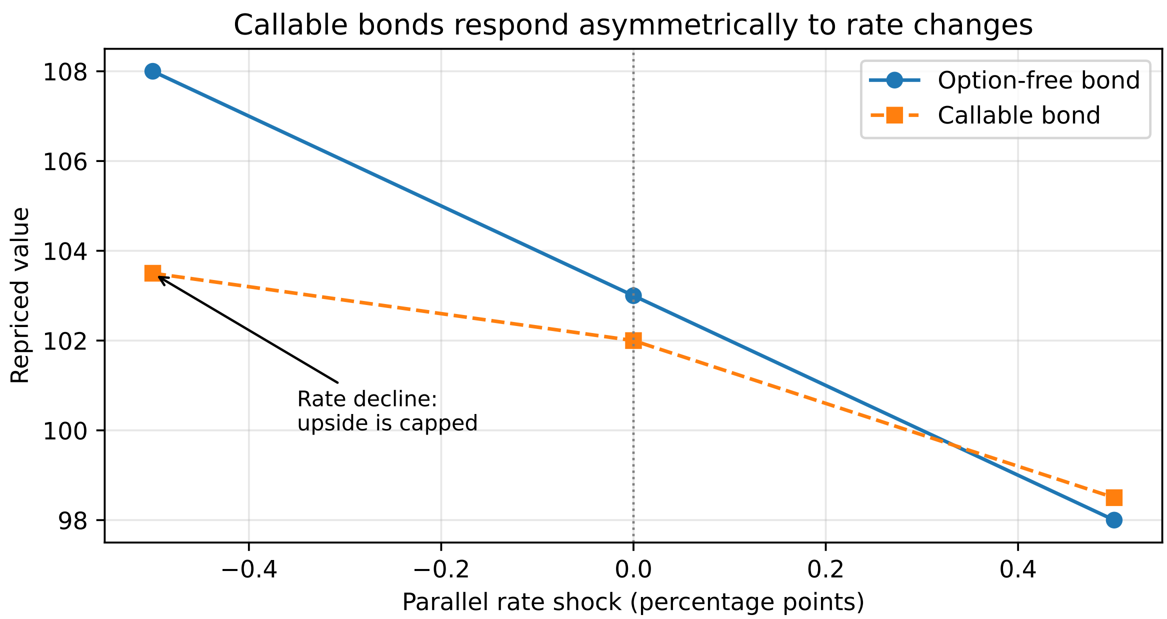

Figure 17: Effective duration and convexity reprice the bond after rate shocks

Key Points

- Embedded options make bond valuation state-dependent: we need future rate states, future bond values, and an exercise rule.
- **Yield spread** is easy to compute but compresses many dated cash flows into one yield and mixes credit, liquidity, curve, and option compensation.
- **Static spread** improves on nominal spread by discounting promised cash flows off the full spot curve, but it still assumes the cash-flow schedule is fixed.
- A **callable bond** is worth less than an otherwise identical option-free bond because the investor is short the issuer's call option; a **putable bond** is worth more because the investor owns the put option.
- Callable bonds have capped upside when rates fall, which can create negative convexity.
- The **interest-rate lattice** is the core valuation tool because it models future short-rate uncertainty and allows exercise decisions at each node.
- **Backward induction** starts from future payoffs and works back to today's price; callable bonds apply a min operator and putable bonds apply a max operator at exercise nodes.
- A model-based tree needs calibration knobs, usually one level parameter per time step, so it can fit the observed zero curve while volatility controls the spacing between states.
- The step-by-step lattice examples show that exercise is state-dependent: the issuer calls when the hold value exceeds the call price, while the investor puts when the put price

exceeds the hold value.

- **OAS** is the spread that equates market price with model price after accounting for option exercise, making it cleaner than nominal or static spread for option-embedded bonds.
- **Effective duration** and **effective convexity** reprice the bond after curve shocks, so they can capture changing cash flows and changing exercise probabilities.

Suggested Readings

Fabozzi, *Bond Markets: Analysis and Strategies*, Chapter 18.